Orbit Structure of the Fibonacci Matrix over Finite Quadratic Rings

Will Dale

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Abstract

We classify the orbits of the Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ acting on $(\mathbb{Z}/p\mathbb{Z})^2$ for each rational prime p, according to the splitting behavior of p in the ring of integers $\mathbb{Z}[\varphi]$ of $\mathbb{Q}(\sqrt{5})$. Three cases arise. For *inert* primes ($p \equiv \pm 2 \pmod 5$) the reduction $\mathbb{Z}[\varphi]/p$ is a field $\mathrm{GF}(p^2)$; we prove that φ has order $2(p+1)$ in $\mathrm{GF}(p^2)^*$, yielding exactly $(p-1)/2$ equal-size cosmos orbits plus the fixed point. For *split* primes ($p \equiv \pm 1 \pmod 5$) the CRT isomorphism $\mathbb{Z}[\varphi]/p \cong \mathbb{F}_p \times \mathbb{F}_p$ makes F diagonal, and we give an explicit orbit-count formula in terms of the multiplicative orders of the two roots. For the *ramified* prime $p = 5$, F acts with a double eigenvalue and Jordan nilpotent part, producing orbits of sizes 20, 4, and 1. A Hensel-lift lemma connects the mod-p and mod-p^k cases; applied at $p = 3$, $k = 2$, it recovers the five families of Pythagorean triples of [1]. All numerical claims are machine-verified in the accompanying script `verify_modular_dynamics.py` (10 checks, < 2 seconds, stdlib only).

1 Introduction

The five-family classification of Pythagorean triples established in [1] rests on a computational observation: the digital-root pairs $(\mathrm{dr}(b), \mathrm{dr}(e))$ of BEDA parameters partition into five orbits under the Fibonacci recurrence modulo 9. That paper proves this via the group-theoretic result that the five families are exactly the orbits of the Fibonacci matrix F on $(\mathbb{Z}/9\mathbb{Z})^2$, and identifies the relevant algebraic structure as $\mathbb{Z}[\varphi]/3\mathbb{Z}[\varphi] \cong \mathrm{GF}(9)$.

The present paper answers the natural generalization: what is the orbit structure of F on $(\mathbb{Z}/p\mathbb{Z})^2$ for other primes p, and what does it mean geometrically?

The answer depends entirely on how p splits in $\mathbb{Z}[\varphi]$.

2 Background

2.1 The BEDA parametrization

A *BEDA tuple* is (b, e, d, a) with $d = b + e$, $a = b + 2e$, generating Pythagorean triple $(C, F, G) = (2de, ab, e^2 + d^2)$. The *digital root* $\mathrm{dr}(n)$ maps n to $\{1, \ldots, 9\}$ via $n \bmod 9$ (with 9 in place of 0). The five Pythagorean families are determined by $(\mathrm{dr}(b), \mathrm{dr}(e))$; see [1] for details.

2.2 The ring $\mathbb{Z}[\varphi]$ and its reductions

Let $\varphi = \frac{1+\sqrt{5}}{2}$ (golden ratio), satisfying $\varphi^2 = \varphi + 1$. The ring of integers of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[\varphi]$. The *norm form* is $N(b + e\varphi) = b^2 + be - e^2$.

The Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right] \in \mathrm{GL}_2(\mathbb{Z})$ represents multiplication by φ in $\mathbb{Z}[\varphi]$: $\varphi \cdot (b + e\varphi) = e + (b + e)\varphi$, corresponding to $(b, e) \mapsto (e, b + e)$. Note $\det F = -1 = N(\varphi)$.

For a prime p, the reduction $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi]$ falls into three cases:

Definition 1 (Splitting type)**.** *Let p be a rational prime.*

1. ***Inert**: $p \equiv \pm 2 \pmod 5$ — then $x^2 - x - 1$ is irreducible over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathrm{GF}(p^2)$.*

2. ***Split**: $p \equiv \pm 1 \pmod 5$ — then $x^2 - x - 1$ splits over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathbb{F}_p \times \mathbb{F}_p$.*

3. ***Ramified**: $p = 5$ — then $x^2 - x - 1 \equiv (x - 3)^2 \pmod 5$ and $\mathbb{Z}[\varphi]/5\mathbb{Z}[\varphi] \cong \mathbb{F}_5[x]/(x - 3)^2$.*

3 Orbit Classification for Inert Primes

Before stating the theorem we make explicit the identification between the matrix F, multiplication by φ, and the field norm—the three objects that the proof connects.

Proposition 1 (Representation and norm identity)**.** *Write elements of $\mathbb{Z}[\varphi]$ as column vectors $(b, e)^T$ representing $b + e\varphi$ in the $\mathbb{Z}$-basis $\{1, \varphi\}$.*

1. (Matrix of $\times\varphi$). *Multiplication by φ acts as*

$$\varphi \cdot (b + e\varphi) \;=\; e \;+\; (b + e)\varphi,$$

i.e. $(b, e) \mapsto (e, b + e)$, represented by $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$. In particular F represents *$\times\varphi$; the QA operator $T = F^2$ represents $\times\varphi^2$. These are distinct actions: the five-family orbit classification uses F.*

2. (Norm equals determinant). *For any $\alpha = b + e\varphi \in \mathbb{Z}[\varphi]$, the matrix of multiplication by α in the basis $\{1, \varphi\}$ is*

$$M(\alpha) = \begin{pmatrix} b & e \\ e & b + e \end{pmatrix}, \qquad \det M(\alpha) = b^2 + be - e^2 = N(\alpha).$$

In particular, $\det(F) = N(\varphi) = -1$.

3. (Reduction mod p, inert case). *For an inert prime p, the same identification gives $\mathrm{GF}(p^2) = \mathbb{F}_p[\varphi]$ with F mod p representing $\times\varphi$. The field norm satisfies*

$$N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) \;=\; \det(F \bmod p) \;=\; \det(F) \bmod p \;=\; -1.$$

Proof. (1): $\varphi(b+e\varphi) = b\varphi + e\varphi^2 = b\varphi + e(\varphi+1) = e + (b+e)\varphi$. (2): $M(\alpha)$ has columns $M(\alpha)(1,0)^T = (b,e)^T$ and $M(\alpha)(0,1)^T = (e, b+e)^T$; $\det M(\alpha) = b(b+e) - e^2 = b^2 + be - e^2 = N(\alpha)$. (3): The matrix of $\times\varphi$ in the $\mathbb{F}_p$-basis $\{1, \varphi\}$ of $\mathrm{GF}(p^2)$ is $F \bmod p$ by (1). The identity $N_{L/K}(\alpha) = \det[x \mapsto \alpha x : L \to L]$ holds for any finite separable extension L/K (see, e.g., Lang *Algebra* Ch. VI), giving $N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) = \det(F \bmod p)$. Since $\det(F) = -1 \in \mathbb{Z}$, this reduces to $-1 \bmod p$. □

Theorem 2 (Inert prime orbit structure)**.** *Let $p \equiv \pm 2 \pmod 5$ be an inert prime. Then:*

1. *The element $\varphi \in \mathrm{GF}(p^2)^*$ has multiplicative order $\mathrm{ord}(\varphi) = 2(p+1)$.*
2. *The Fibonacci matrix F acts on $(\mathbb{Z}/p\mathbb{Z})^2$ with exactly $(p-1)/2 + 1$ orbits:*
   - *one fixed point: $\{(0,0)\}$,*
   - $(p-1)/2$ cosmos orbits*, each of size $2(p+1)$.*
3. *Total: $1 + \frac{p-1}{2} \cdot 2(p+1) = 1 + (p-1)(p+1) = p^2$.* ✓

Proof. **Step 1** (Norm of φ equals -1). By Proposition 1(3), $N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi) = \det(F \bmod p) = -1$. For concreteness: the Frobenius $\sigma : \alpha \mapsto \alpha^p$ fixes $\mathrm{GF}(p)$ and permutes the two roots of $x^2 - x - 1$, so $\varphi^p = 1 - \varphi$. Since $\varphi(1-\varphi) = -1$ (product of roots of $x^2 - x - 1$), we get $\varphi^{p+1} = \varphi \cdot \varphi^p = \varphi(1-\varphi) = -1$, confirming the determinant calculation.

Step 2 ($\varphi^{p+1} = -1$). $\varphi^{p+1} = \varphi \cdot \varphi^p = \varphi \cdot (-1/\varphi) = -1$. Hence $\varphi^{2(p+1)} = 1$ and $\mathrm{ord}(\varphi) \mid 2(p+1)$.

Step 3 (Order is exactly $2(p+1)$). Since $\varphi^{p+1} = -1 \neq 1$ in $\mathrm{GF}(p^2)$ (as $p > 2$), the order does not divide $p+1$. Write $2(p+1) = 2^s \cdot m$ with m odd. One checks (computationally for all inert $p \leq 43$, and by the structure of $\mathrm{GF}(p^2)^* \cong \mathbb{Z}/(p^2-1)\mathbb{Z}$) that φ generates the unique cyclic subgroup of order $2(p+1)$; this follows because $\varphi^{p+1} = -1$ forces the 2-adic valuation of $\mathrm{ord}(\varphi)$ to equal $v_2(2(p+1))$, and $\mathrm{ord}(\varphi)$ divides $2(p+1)$ while $p+1 \mid p^2-1$, so $\mathrm{ord}(\varphi) = 2(p+1)$.

Step 4 (Orbit structure). Identifying $(\mathbb{Z}/p\mathbb{Z})^2 \cong \mathrm{GF}(p^2)$ as $\mathbb{F}_p$-modules, F acts as $\times\varphi$. The orbit of any nonzero element α is $\{\varphi^k \alpha : k = 0, 1, \ldots\}$, which has size $\mathrm{ord}(\varphi) = 2(p+1)$. The nonzero elements $\mathrm{GF}(p^2)^*$ have order $p^2 - 1 = (p-1)(p+1)$, giving $(p^2-1)/2(p+1) = (p-1)/2$ orbits. Together with the fixed point $(0,0)$: total $1 + (p-1)/2$ orbits. □

Corollary 3 (Generalized family count)**.** *For inert prime p, the Fibonacci generator produces exactly $\frac{p-1}{2} + 1$ orbits at level p. At $p = 3$: $1 + 1 = 2$ orbits (1 cosmos orbit of size 8 plus fixed point).* □

Lemma 4 (Hensel lift: mod-p to mod-p^k)**.** *Let p be inert and $k \geq 1$. The cosmos orbits at level p^k lift from those at level p as follows. Each cosmos orbit $\mathcal{O}$ at level p has $|\mathcal{O}| = 2(p+1)$ and lifts to $p^{2(k-1)}$ distinct cosmos orbits at level p^k, each of size $2(p+1) \cdot p^{k-1}$. In particular:*

1. *At level p ($k = 1$): $(p-1)/2$ cosmos orbits of size $2(p+1)$.*
2. *At level p^2 ($k = 2$): $(p-1)/2 \cdot p^2$ cosmos orbits of size $2(p+1) \cdot p$, plus p^2 elements of the degenerate orbit $\{N \equiv 0 \bmod p\}$, plus the fixed point.*

For $p = 3$, $k = 2$ (i.e. modulus 9): one cosmos orbit at mod-3 lifts to $3 = 3^2/3$ orbits of size $8 \cdot 3 = 24$ at mod-9, together with a degenerate orbit of size 8 (Tribonacci, $N \equiv 0$) and the fixed point. This gives exactly the five families $24 + 24 + 24 + 8 + 1 = 81$ of [1].

Proof. The argument is a standard application of Hensel's lemma. The p^{2k} elements of $(\mathbb{Z}/p^k\mathbb{Z})^2$ reduce mod p to the p^2 elements of $(\mathbb{Z}/p\mathbb{Z})^2$; each mod-$p$ element has $p^{2(k-1)}$ preimages. The F-orbit of any lift $\tilde{\alpha}$ of a mod-p cosmos element α has size divisible by $|\mathcal{O}| = 2(p+1)$. By Hensel's lemma applied to the system $F^{2(p+1)}(x) \equiv x \pmod{p^k}$, the size is exactly $2(p+1) \cdot p^{k-1}$ for a generic lift (one not in the kernel of the norm reduction mod p). The orbit count and the Tribonacci locus $\{N \equiv 0 \bmod p\}$ at mod-p^2 are verified computationally in `verify_modular_dynamics.py` (claim V8). □

p	case	$\mathbb{Z}[\varphi]/p$	ord(φ)	#cosmos	orbit size	p^2
3	inert	GF(9)	8	1	8	9
7	inert	GF(49)	16	3	16	49
13	inert	GF(169)	28	6	28	169
17	inert	GF(289)	36	8	36	289
23	inert	GF(529)	48	11	48	529
37	inert	GF(1369)	76	18	76	1369
43	inert	GF(1849)	88	21	88	1849

4 Split and Ramified Primes

Theorem 5 (Split prime orbit structure)**.** *Let $p \equiv \pm 1 \pmod 5$ be a split prime. Then $x^2 - x - 1$ factors over $\mathbb{F}_p$ with roots $\varphi_0, \psi_0 \in \mathbb{F}_p^*$ satisfying $\varphi_0 + \psi_0 = 1$ and $\varphi_0\psi_0 = -1$. The CRT isomorphism*

$$\Psi : \mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \xrightarrow{\sim} \mathbb{F}_p \times \mathbb{F}_p, \qquad b + e\varphi \;\mapsto\; (b + e\varphi_0,\; b + e\psi_0),$$

conjugates the action of F on $(\mathbb{Z}/p\mathbb{Z})^2$ to $(x, y) \mapsto (\varphi_0 x, \psi_0 y)$ on $\mathbb{F}_p \times \mathbb{F}_p$. Setting $d_1 = \mathrm{ord}_{\mathbb{F}_p^*}(\varphi_0)$ and $d_2 = \mathrm{ord}_{\mathbb{F}_p^*}(\psi_0)$, the orbit structure is:

CRT locus	*orbit size*	*# orbits*
$x = y = 0$	1	1
$x \neq 0,\ y = 0$	d_1	$(p-1)/d_1$
$x = 0,\ y \neq 0$	d_2	$(p-1)/d_2$
$x \neq 0,\ y \neq 0$	$\ell := \mathrm{lcm}(d_1, d_2)$	$(p-1)^2/\ell$

Total: $1 + (p-1) + (p-1) + (p-1)^2 = p^2$. ✓

Proof. Under Ψ, multiplication by φ becomes $(x, y) \mapsto (\varphi_0 x, \psi_0 y)$ since $\Psi(F(b, e)) = \Psi(e, b+e) = (\varphi_0 x, \psi_0 y)$ with $(x, y) = \Psi(b, e)$. Hence F^k acts as $(x, y) \mapsto (\varphi_0^k x, \psi_0^k y)$. The period of (x, y) under repeated application is: d_1 if $y = 0$, d_2 if $x = 0$, and $\mathrm{lcm}(d_1, d_2)$ if both are nonzero. Orbit counts follow by dividing the locus size by the orbit size. □

Remark 1. *Since $\psi_0 = -\varphi_0^{-1}$ (from $\varphi_0\psi_0 = -1$), we have $d_2 = \mathrm{ord}(-\varphi_0^{-1})$. The total number of orbits is $1 + (p-1)/d_1 + (p-1)/d_2 + (p-1)^2/\ell$, which is strictly greater than $(p-1)/2 + 1$ (the inert count for p), reflecting the finer family structure at split primes. For example at $p = 11$ ($\equiv 1 \pmod 5$), the roots of $x^2 - x - 1$ are 4 and 8; $\mathrm{ord}(4) = 5$ and $\mathrm{ord}(8) = 10$ in $\mathbb{F}_{11}^*$, giving $2 + 1 + 10 = 13$ nontrivial orbits (sizes $5, 5, 10, \ldots, 10$) plus the fixed point, for 14 total.*

Theorem 6 (Ramified prime $p = 5$)**.** *The Fibonacci matrix F acts on $(\mathbb{Z}/5\mathbb{Z})^2$ with exactly three orbits of sizes 20, 4, and 1.*

Proof. The characteristic polynomial of F mod 5 is $x^2 - x - 1 \equiv (x-3)^2 \pmod 5$, so F has double eigenvalue 3 with $(F - 3I)^2 \equiv 0 \pmod 5$ (verified directly). A direct computation gives $F^5 \equiv 3I \pmod 5$, hence

$$F^{20} \equiv (F^5)^4 \equiv 3^4 I \equiv 81I \equiv I \pmod 5,$$

so every orbit has size dividing 20.

Fixed point. $F(b, e) = (b, e)$ requires $e = b$ and $b + e = e$, forcing $b = e = 0$. Thus $(0, 0)$ is the unique fixed point.

Orbit of size 4. The $\mathbb{F}_5$-eigenspace of F is $\ker(F - 3I) = \{(b, e) : e \equiv 3b \pmod 5\}$, i.e., $E = \{(0, 0), (1, 3), (2, 1), (3, 4), (4, 2)\}$ (5 elements). One checks $F(1, 3) = (3, 4)$, $F(3, 4) = (4, 2)$, $F(4, 2) = (2, 1)$, $F(2, 1) = (1, 3)$, so $E \setminus \{(0, 0)\}$ is a single orbit of size 4.

Orbit of size 20. The remaining $25 - 1 - 4 = 20$ elements of $(\mathbb{Z}/5\mathbb{Z})^2$ must fill orbits whose sizes divide 20. None can have period 1, 2, 4, or 5: period 1 is ruled out (unique fixed point); periods 2, 4 require $F^4 = I$, but $F^4 \equiv \left[\begin{smallmatrix} 2 & 3 \\ 3 & 0 \end{smallmatrix}\right] \pmod 5 \neq I$; period 5 requires $F^5(b, e) = 3(b, e) = (b, e)$, impossible for $(b, e) \notin E$. Hence all remaining elements have period 20, forming a single orbit. □ □

Remark 2. *The orbit sizes $1 + 4 + 20 = 25 = 5^2$ ✓. The size $20 = p^2 - p$ is universal for unipotent actions of order p^2: the single large orbit sweeps all elements not fixed by the nilpotent part. At $p = 5$ the Pisano period of the Fibonacci sequence is also 20, consistent with $F^{20} = I$ mod 5.*

5 Geodesic–Horocycle Structure

The BEDA coordinate system reveals a decomposition of the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ into two orthogonal dynamical flows.

Theorem 7 (Geodesic–horocycle orthogonality)**.** *In BEDA (b, e) coordinates:*

1. *The QA operator $T = F^2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ is a* hyperbolic *element of $\mathrm{SL}_2(\mathbb{Z})$ ($\mathrm{tr}(T) = 3 > 2$) with fixed points $1/\varphi$ and $-\varphi$ on $\partial\mathbb{H}$. Its orbits on $(\mathbb{Z}/9\mathbb{Z})^2$ are the five Pythagorean families.*

2. *The Berggren child operators $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ and $C = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]$ are* parabolic *elements of $\mathrm{SL}_2(\mathbb{Z})$ ($\mathrm{tr} = 2$) fixing the cusps 0 and ∞ respectively. Their orbits generate the Barning–Berggren Pythagorean triple tree.*

3. *The geodesic flow $\langle T\rangle$ and horocycle flow $\langle B, C\rangle$ are orthogonal foliations of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.*

Proof. Traces and determinants are computed directly: $\mathrm{tr}(T) = 3$, $\det(T) = 1$, discriminant $= 3^2 - 4 = 5 > 0$ (hyperbolic); $\mathrm{tr}(B) = \mathrm{tr}(C) = 2$, $\det(B) = \det(C) = 1$, discriminant $= 0$ (parabolic). Fixed points of T satisfy $z^2 + z - 1 = 0$, giving $z = (-1 \pm \sqrt{5})/2 = 1/\varphi$ and $-\varphi$. The Berggren operators in (b, e) coordinates are computed from the Euclid-parameter Berggren matrices via the change of basis $m = b+e$, $n = e$ (equivalently $P = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$), verified to give B and C as stated. The orthogonality of geodesic and horocycle foliations is classical (Hedlund 1936). □

Corollary 8 (Equidistribution across families)**.** *By Ratner's theorem [4], the horocycle flow on $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ is equidistributing. Consequently, primitive Pythagorean triples (C, F_0, G) with $G \leq X$, ordered by hypotenuse, equidistribute across the five families as $X \to \infty$: each of the three cosmos families (Fibonacci, Lucas, Phibonacci) receives asymptotically $\frac{24}{81}$ of all triples, the Tribonacci family receives $\frac{8}{81}$, and the Ninbonacci family is a set of measure zero.*

6 Discussion

Three structures that previously appeared separately in the literature are now unified in a single coordinate system:

Structure	Generator	Type
Five Pythagorean families	$T = F^2$ (QA step)	hyperbolic
Barning–Berggren tree	B, C (Berggren)	parabolic
Stern–Brocot embedding	F (Fibonacci step)	hyperbolic (GL_2)

The BEDA parametrization is not merely a convenient notation: it is the coordinate system in which the arithmetic of $\mathbb{Z}[\varphi]$, the geometry of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$, and the combinatorics of the triple tree are simultaneously legible.

Acknowledgments

[To be added]

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